IGCSE 0607 Practice Questions — E1.12

Percentages

Total Questions: 12 — **Total Marks:** 69 Marks

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1 A shopkeeper buys a television for \$320. He sells it for \$400.
Calculate his percentage profit.

..... % **[2]**

2 **(a)** Write 24 as a percentage of 60.

..... % **[2]**

(b) A coat has a normal price of \$80. In a sale, the price is reduced by 15%.
Work out the sale price of the coat.

\$ ..... **[2]**

3 **(a)** Find 15% of \$320.

\$ ..... **[2]**

(b) In a sale, the original price of a shirt is reduced by 20%.
The original price of the shirt is \$45.
Work out the sale price.

\$ ..... **[3]**

4 A shop sells a jacket for \$120. During a sale, the shop offers a 15% discount on the price of the jacket.

(a) Calculate the sale price of the jacket.

\$ ..... **[2]**

(b) The original price of \$120 gives the shop a profit of 20% on the cost price.
Calculate the cost price of the jacket.

\$ ..... **[3]**

5 **(a)** Alice invests \$4500 at a rate of 3.2% per year compound interest.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

(b) A shopkeeper buys a television for \$320. He sells it for \$408.
Calculate his percentage profit.

..... % **[2]**

6 **(a)** In 2021, the population of a town was 45 000.
In 2022, the population increased by 4.2%.
Calculate the population in 2022.

..... **[2]**

(b) In a sale, the price of a television is reduced by 24%.
The sale price is $342.
Calculate the original price of the television.

$ ..... **[3]**

7 Amira invests \$4500 in a savings account that pays compound interest at a rate of 2.4% per year.

(a) Calculate the total amount in Amira's account at the end of 5 years.

\$ ..... **[3]**

(b) Ben invests \$4500 in a different savings account that pays simple interest at a rate of r% per year. At the end of 5 years, Ben has the same total amount in his account as Amira has in hers.

Calculate the value of r.

$r =$ ..... **[3]**

8 **(a)** Amira invests \$4500 in a savings account paying compound interest at a rate of 2.4% per year.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

(b) In a sale, the price of a computer is reduced by 18%.
The sale price is \$557.60.
Calculate the original price of the computer.

\$ ..... **[3]**

9 **(a)** The population of a city increases by 4.5% each year.
The population is now 1.25×10^6.
Calculate the expected population in 10 years' time.
Give your answer in standard form, correct to 3 significant figures.

..... **[3]**

(b) In a sale, a washing machine is sold for \$299.25. This is a 14% reduction on the original price.
Calculate the original price.

\$ ..... **[2]**

(c) A car depreciates in value by 15% in the first year and by 10% in each subsequent year.
The value of the car after 3 years is \$17212.50.
Calculate the original value of the car.

\$ ..... **[2]**

10 The population of a city increases by $x\%$ each year.
In 2015, the population was 1 250 000.
In 2018, the population was 1 367 631.

(a) Show that the value of x is approximately 3.

[3]

(b) Assume the population continues to increase by exactly 3% each year.
Calculate the year in which the population will first exceed double its 2015 value.

..... **[4]**

11 **(a)** Ali invests \$5000 at a rate of 2.5% per year compound interest. Calculate the value of his investment at the end of 3 years.
Give your answer correct to the nearest cent.

\$ ..... **[2]**

(b) Beatrice invests a sum of money at a rate of r% per year compound interest.
After 5 years, the value of her investment has increased by 30%.
Calculate the value of r.

$r =$ ..... **[3]**

(c) The value of a car depreciates by 12% each year.
After 3 years, the value of the car is \$17 036.80.
Calculate the original value of the car.

\$ ..... **[3]**

12 **(a)** A car's value depreciates by 15% in the first year.
In each subsequent year, its value depreciates by 8%.
After exactly 3 years, the value of the car is \$17 986.
Calculate the original value of the car.

\$ ..... **[4]**

(b) Calculate the overall percentage decrease in the car's value over the 3 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

(c) The population of a city is P.
The population increases by r% each year.
After 12 years, the population has exactly doubled.
Calculate the value of r.

$r =$ ..... **[3]**